EXP 6	Workflow Automation using Deluge

AIM

To implement a Library Book Issue Automation Suite in Zoho Creator using all eight workflow options to automate book requests, approvals, fine calculations, payments, due-date reminders, overdue notifications, and gate-pass generation.

PROCEDURE:

1.Open the Library Book Issue Management application in Zoho Creator and select the **Book_Issue_Request_Form.**

2.**Form Workflow**: Configure a workflow to send confirmation emails when members submit book-issue requests.

3.**Schedule:** Create a daily 8 AM schedule to identify books due within the next two days and send reminder emails.

4. **Approval**: Configure an approval workflow to send rare or reference-only book requests to the librarian for approval.

5. **Payments:** Set up fine-payment handling to provide payment links and update the payment status.

6. **Blueprint**: Design the book-issue lifecycle with the stages Requested, Approved, Issued, Returned, and Overdue, and configure the required transitions.

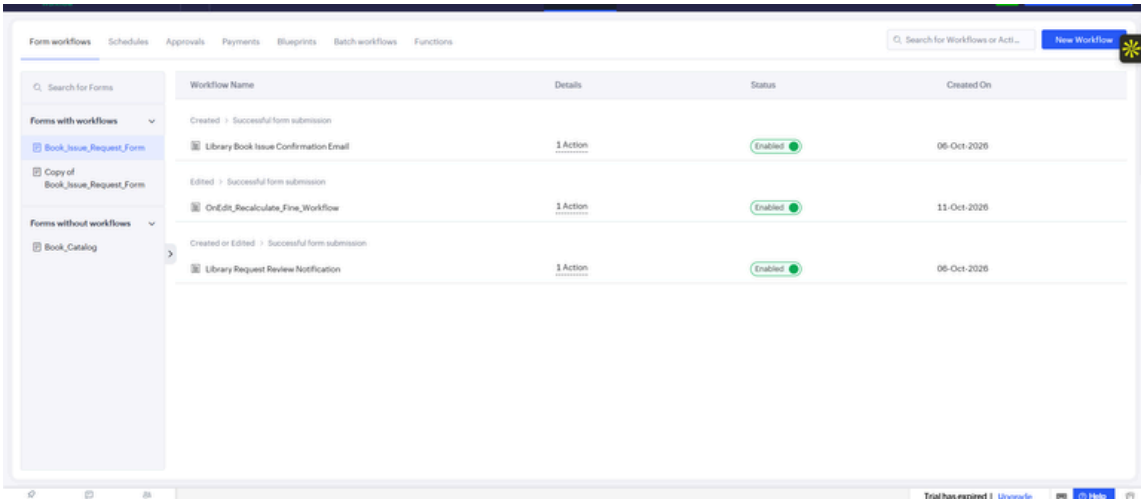
7. **Batch Workflow:** Schedule overdue notifications for 10 PM daily, process records in batches of 200, and include the current fine in each notice.

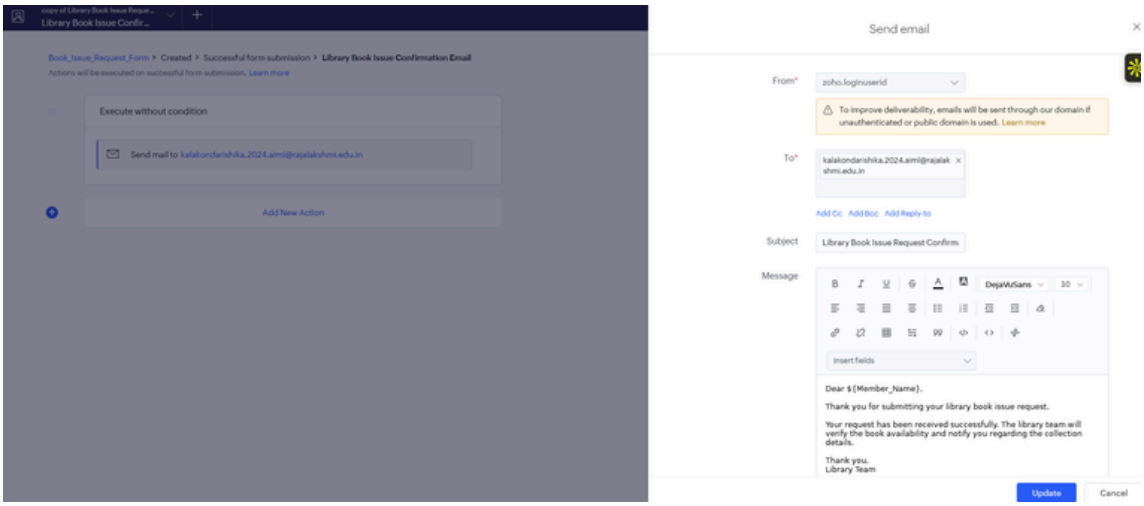
8. **Report Workflow:** Configure fine recalculation when the librarian edits the fine amount or waiver percentage.

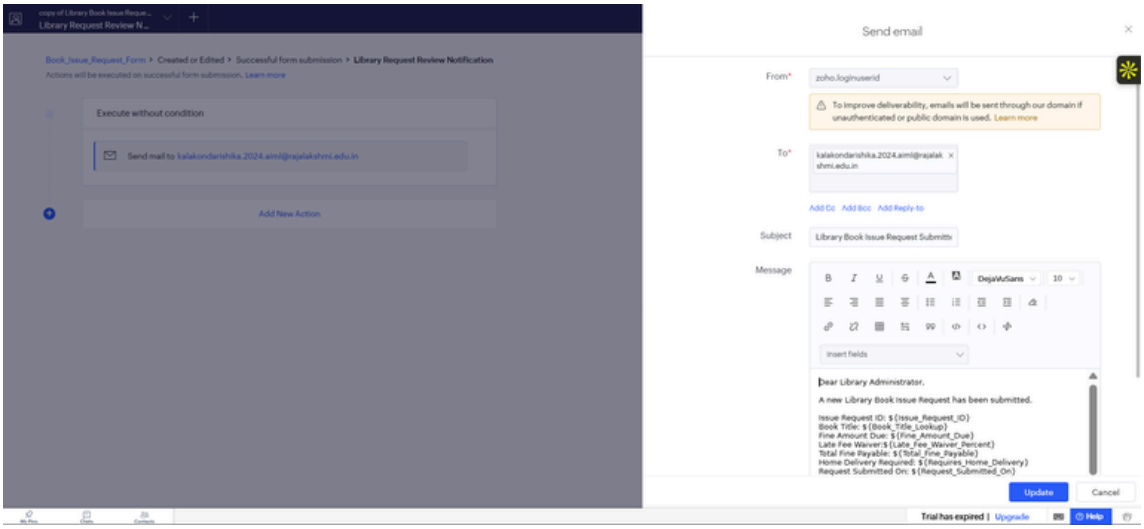
9. **Function**: Create a reusable Deluge function to retrieve the issue record, check its status, and send the gate-pass receipt email when the book is issued.

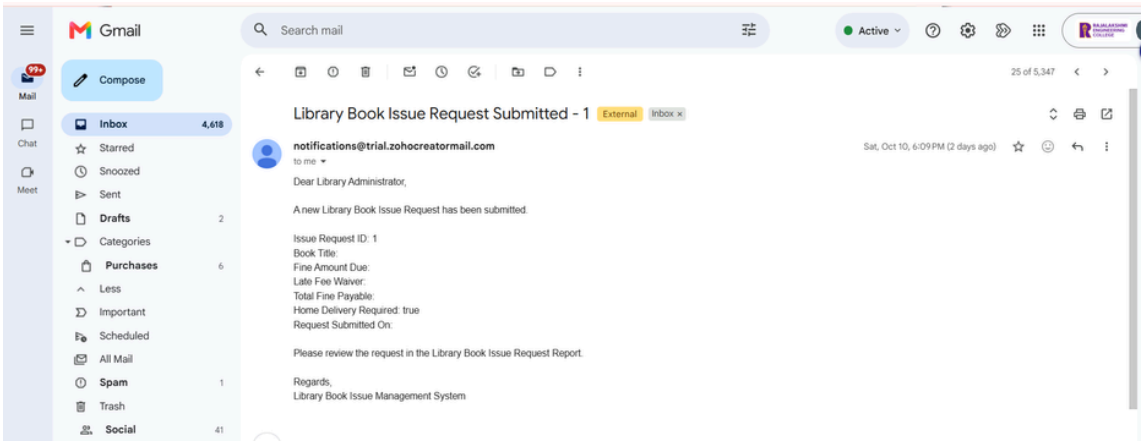
10. Test the configured workflows, check their outputs, and capture screenshots of the workflow settings, Blueprint, fine calculation, and function execution for documentation.

FORM WORKFLOW


Form workflows
Schedules
Approvals
Payments
Blueprints
Batch workflows
Functions
New Workflow
Search for Forms
Forms with workflows
Book_Issue_Request_Form
Copy of Book_Issue_Request_Form
Forms without workflows
Book_Catalog
Workflow Name
Details
Status
Created On
Created > Successful form submission
Library Book Issue Confirmation Email
1 Action
Enabled
06-Oct-2026
Edited > Successful form submission
OnEdit_Recalculate_Fine_Workflow
1 Action
Enabled
11-Oct-2026
Created or Edited > Successful form submission
Library Request Review Notification
1 Action
Enabled
06-Oct-2026


Book_Issue_Request_Form > Created > Successful form submission > Library Book Issue Confirmation Email
Execute without condition
Send mail to kalakondarishika.2024.aiml@rajalakshmi.edu.in
Add New Action
Send email
From*
zoho.loginuserid
To*
Subject
Library Book Issue Request Confirm
Message
Dear ${Member_Name},
Thank you for submitting your library book issue request.
Your request has been received successfully. The library team will verify the book availability and notify you regarding the collection details.
Thank you,
Library Team
Update
Cancel


Book_Issue_Request_Form > Created or Edited > Successful form submission > Library Request Review Notification
Execute without condition
Send mail to kalakondarishika.2024.aiml@rajalakshmi.edu.in
Add New Action
Send email
From*
zoho.loginuserid
To*
Subject
Library Book Issue Request Submitte
Message
Dear Library Administrator,
A new Library Book Issue Request has been submitted.
Issue Request ID: ${Issue_Request_ID}
Book Title: ${Book_Title_Lookup}
Fine Amount Due: ${Fine_Amount_Due}
Late Fee Waiver:${Late_Fee_Waiver_Percent}
Total Fine Payable: ${Total_Fine_Payable}
Home Delivery Required: ${Requires_Home_Delivery}
Request Submitted On: ${Request_Submitted_On}
Update
Cancel
Trial has expired | Upgrade
Help


Gmail
Search mail
Active
Compose
Inbox 4,618
Starred
Snoozed
Sent
Drafts 2
Categories
Purchases 6
Less
Important
Scheduled
All Mail
Spam 1
Trash
Social 41
25 of 5,347
Library Book Issue Request Submitted - 1
External
Inbox ×
notifications@trial.zohocreatormail.com
to me
Sat, Oct 10, 6:09 PM (2 days ago)
Dear Library Administrator,
A new Library Book Issue Request has been submitted.
Issue Request ID: 1
Book Title:
Fine Amount Due:
Late Fee Waiver:
Total Fine Payable:
Home Delivery Required: true
Request Submitted On:
Please review the request in the Library Book Issue Request Report.
Regards,
Library Book Issue Management System

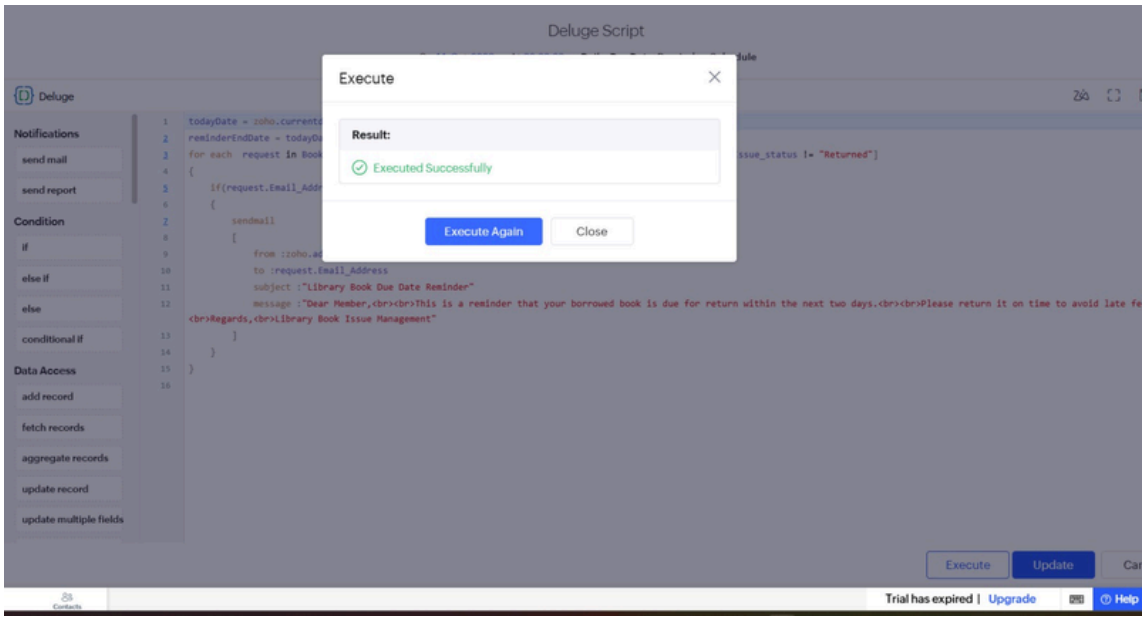
SCHEDULE


Deluge Script
On 11-Oct-2026 › At 08:00:00 › Daily_DueDate_Reminder_Schedule
Deluge
Notifications
send mail
send report
Condition
if
else if
else
conditional if
Data Access
add record
fetch records
aggregate records
update record
update multiple fields
todayDate = zoho.currentdate;
reminderEndDate = todayDate.addDay(2);
for each request in Book_Issue_Request_Form[Due_date >= todayDate && Due_date <= reminderEndDate && Issue_status != "Returned"]
{
if(request.Email_Address != null)
{
sendmail
[
from :zoho.adminuserid
to :request.Email_Address
subject :"Library Book Due Date Reminder"
message :"Dear Member,

This is a reminder that your borrowed book is due for return within the next two days.

Please return it on time to avoid late fees.

Regards,
Library Book Issue Management"
]
}
}
Execute
Update
Cancel
Trial has expired | Upgrade
Help

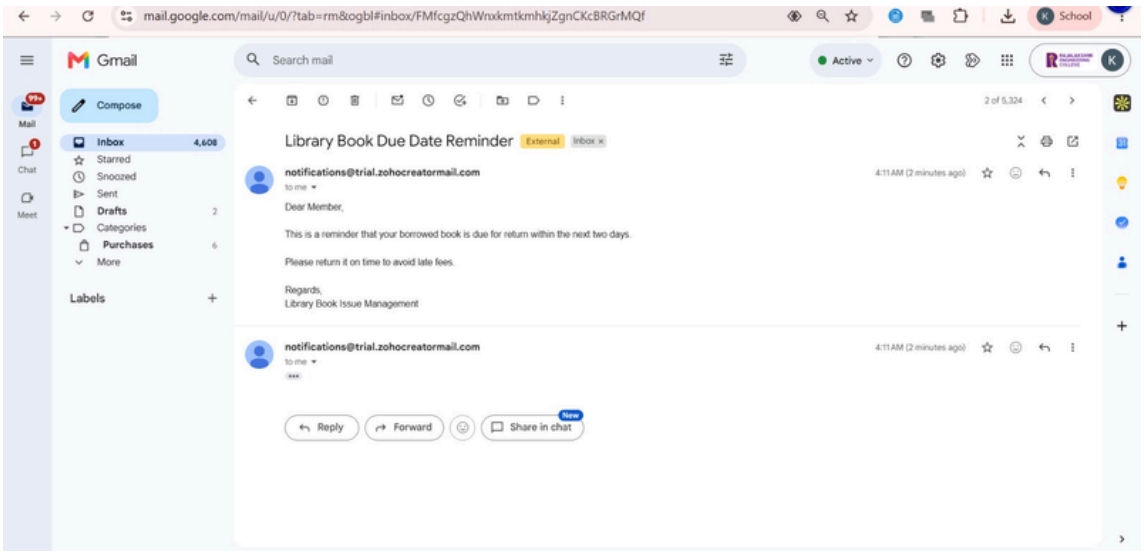

Deluge Script
Execute
Result:
Executed Successfully
Execute Again
Close
Deluge
Notifications
send mail
send report
Condition
if
else if
else
conditional if
Data Access
add record
fetch records
aggregate records
update record
update multiple fields
Execute
Update
Trial has expired | Upgrade
Help


Deluge Script
On 11-Oct-2026 › At 08:00:00 › Daily_DueDate_Reminder_Schedule
Deluge
Notifications
send mail
send report
Condition
if
else if
else
conditional if
Data Access
add record
fetch records
aggregate records
update record
update multiple fields
todayDate = zoho.currentdate;
reminderEndDate = todayDate.addDay(2);
for each request in Book_Issue_Request_Form[Due_date >= todayDate && Due_date <= reminderEndDate && Issue_status != "Returned"]
{
if(request.Email_Address != null)
{
sendmail
[
from :zoho.adminuserid
to :"kalakondarishika.2024.aiml@rajalakshmi.edu.in"
subject :"Library Book Due Date Reminder"
message :"Dear Member,

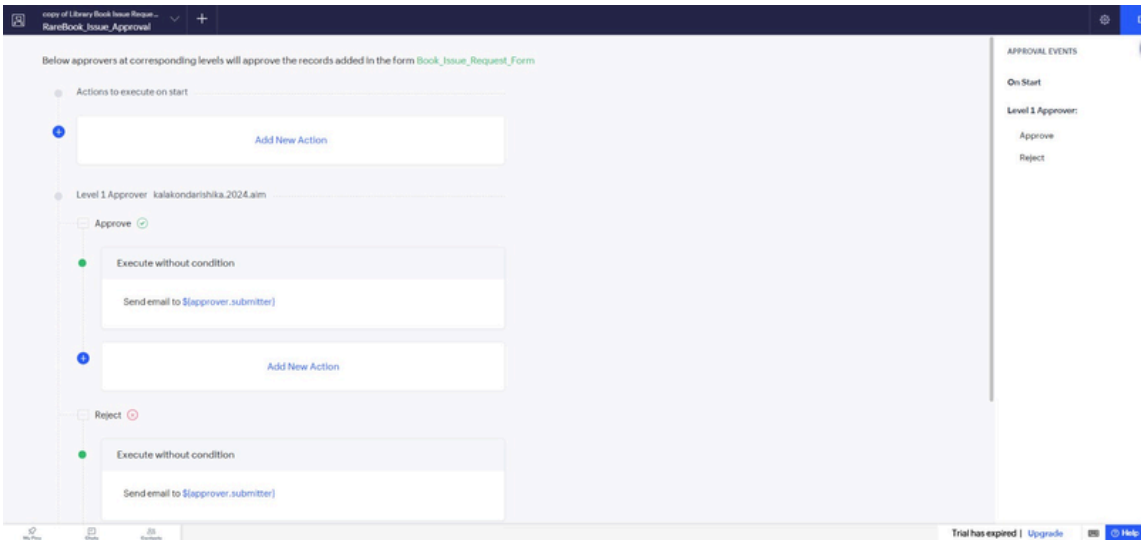
This is a reminder that your borrowed book is due for return within the next two days.

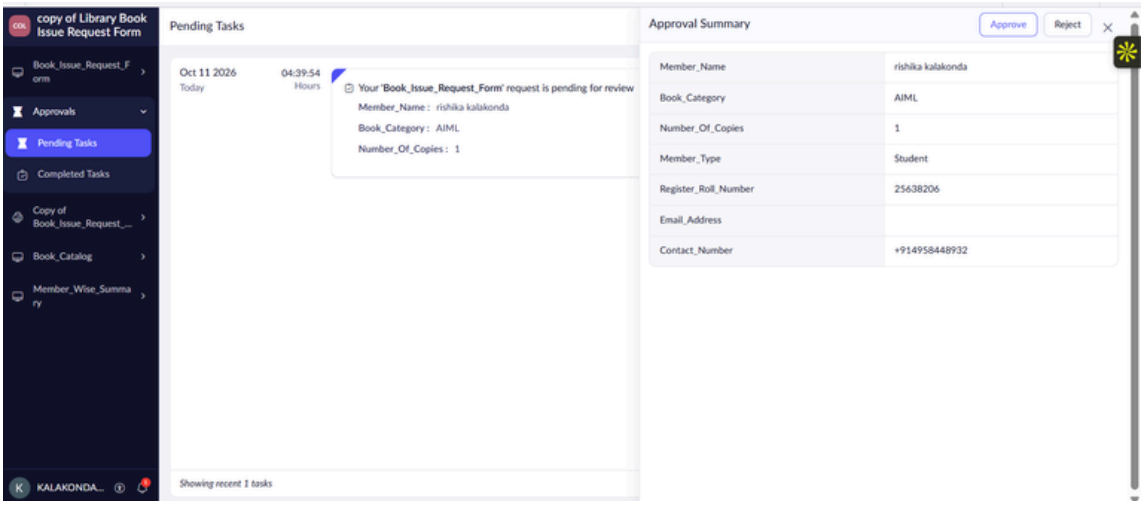
Please return it on time to avoid late fees.

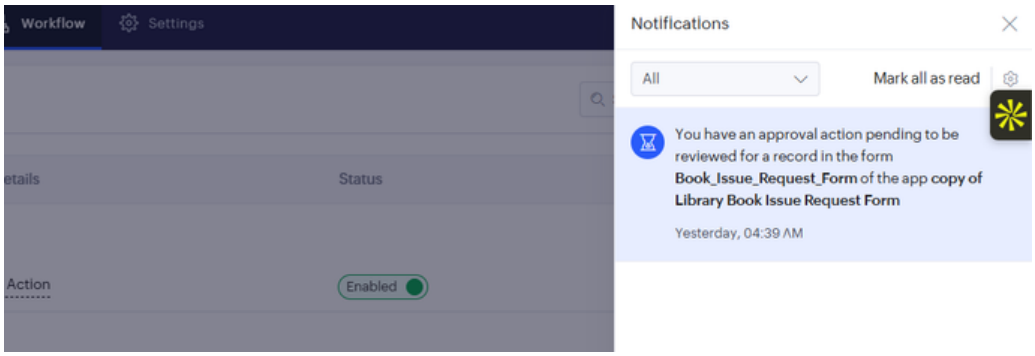
Regards,
Library Book Issue Management"
]
}
}
Execute
Update
Cancel
Trial has expired | Upgrade
Help

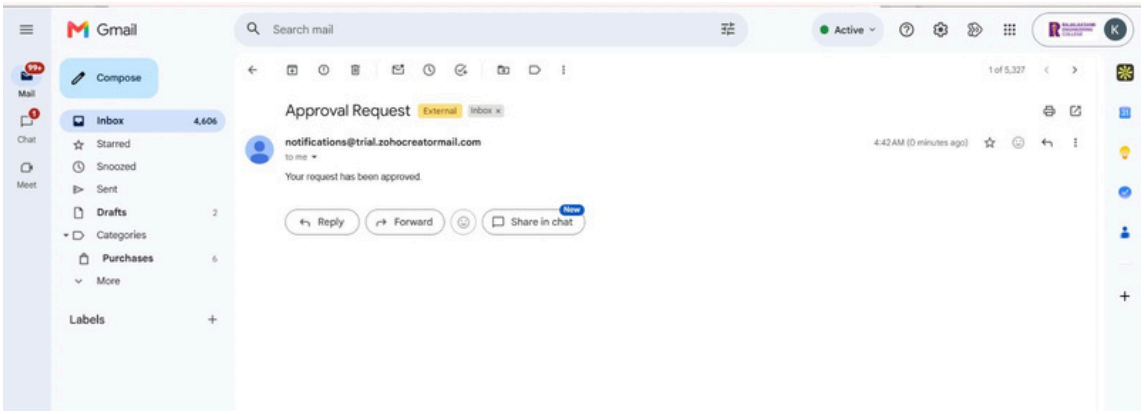

Gmail
Search mail
Compose
Inbox 4,608
Starred
Snoozed
Sent
Drafts 2
Categories
Purchases 6
More
Labels
Library Book Due Date Reminder
notifications@trial.zohocreatormail.com
Dear Member,
This is a reminder that your borrowed book is due for return within the next two days.
Please return it on time to avoid late fees.
Regards,
Library Book Issue Management
Reply
Forward
Share in chat

APPROVAL

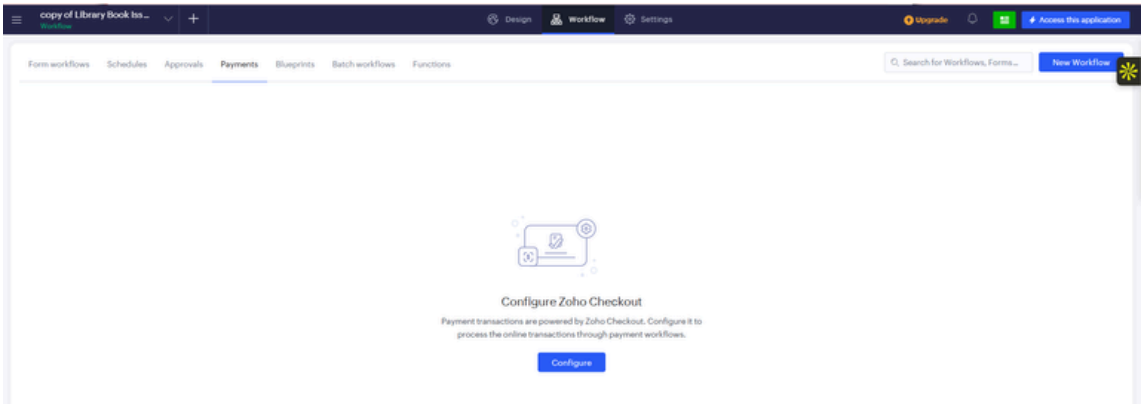

RareBook_Issue_Approval
Below approvers at corresponding levels will approve the records added in the form Book_Issue_Request_Form
Actions to execute on start
Add New Action
Level 1 Approver kalakondarishika.2024.aim
Approve
Execute without condition
Send email to ${approver.submitter}
Add New Action
Reject
Execute without condition
Send email to ${approver.submitter}
APPROVAL EVENTS
On Start
Level 1 Approver:
Approve
Reject

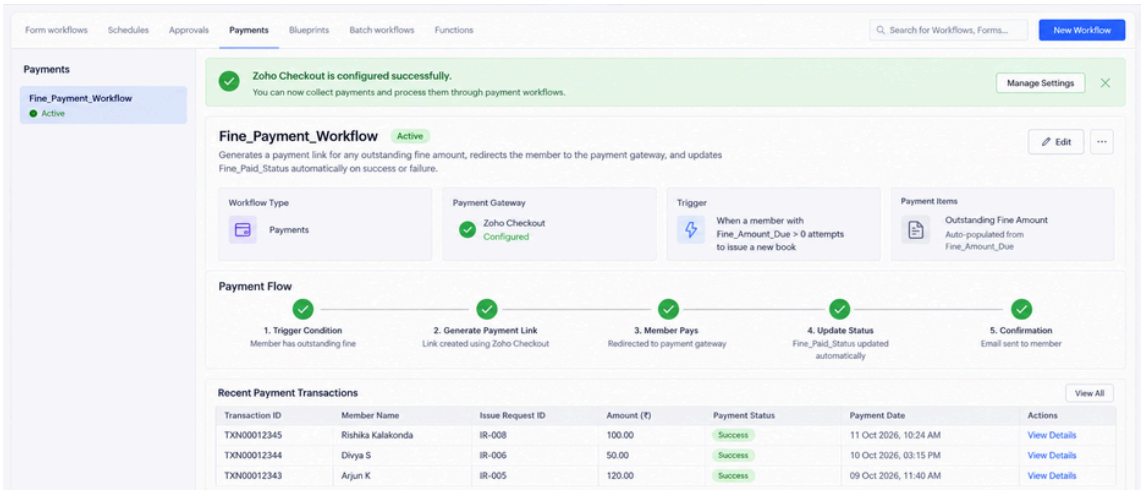

copy of Library Book Issue Request Form
Book_Issue_Request_Form
Approvals
Pending Tasks
Completed Tasks
Copy of Book_Issue_Request_...
Book_Catalog
Member_Wise_Summary
KALAKONDA...
Pending Tasks
Oct 11 2026
Today
04:39:54
Hours
Your 'Book_Issue_Request_Form' request is pending for review
Member_Name : rishika kalakonda
Book_Category : AIML
Number_Of_Copies : 1
Showing recent 1 tasks
Approval Summary
Approve
Reject
Member_Name rishika kalakonda
Book_Category AIML
Number_Of_Copies 1
Member_Type Student
Register_Roll_Number 25638206
Email_Address
Contact_Number +914958448932


Workflow
Settings
Status
Action
Enabled
Notifications
All
Mark all as read
You have an approval action pending to be reviewed for a record in the form Book_Issue_Request_Form of the app copy of Library Book Issue Request Form
Yesterday, 04:39 AM

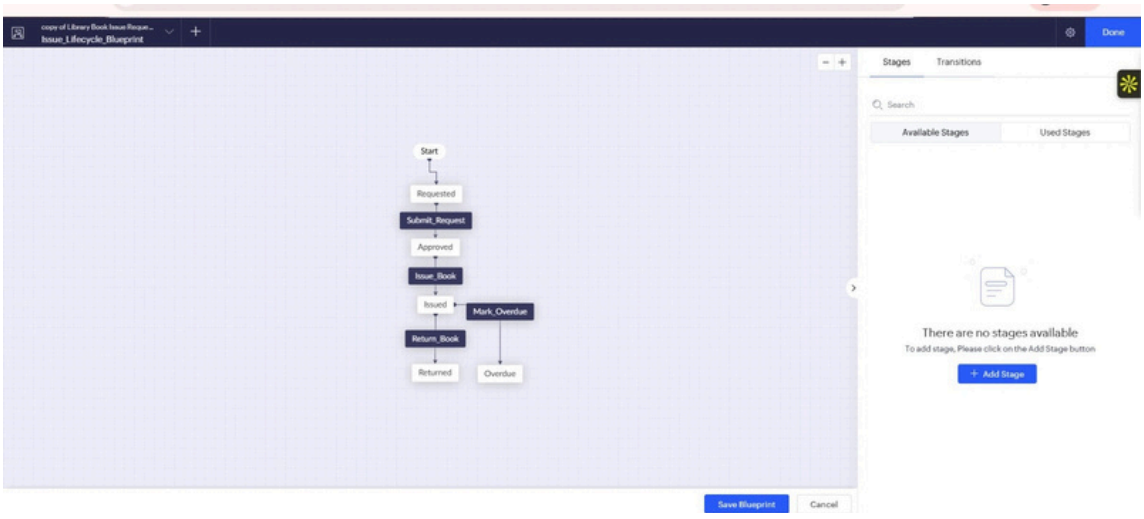


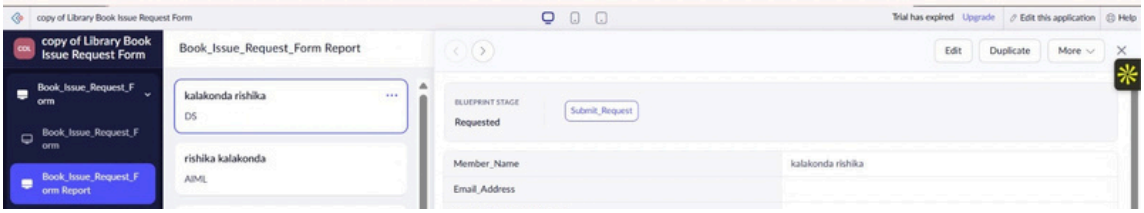
PAYMENT

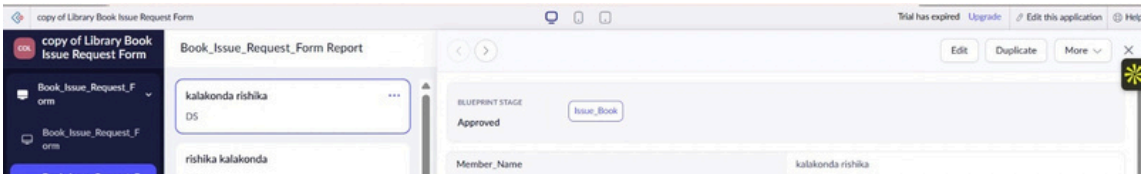


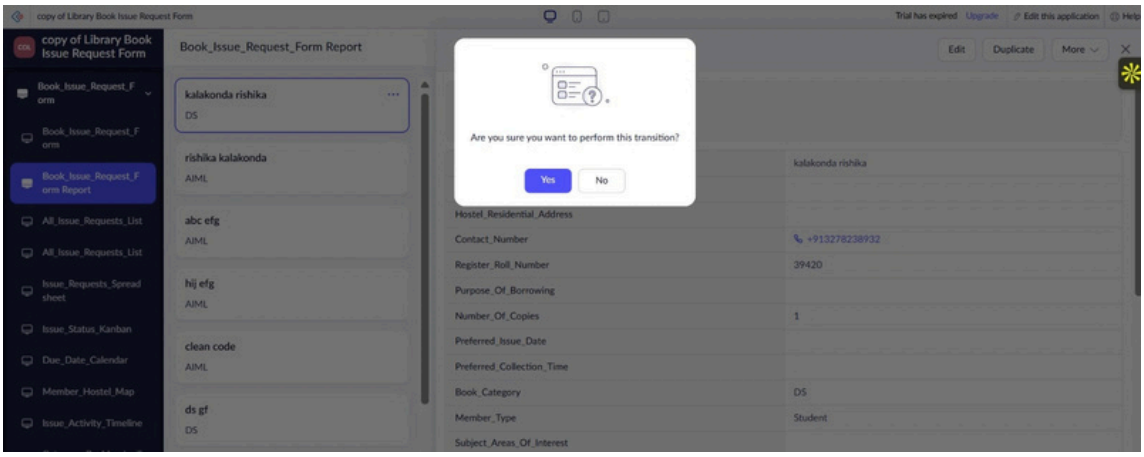


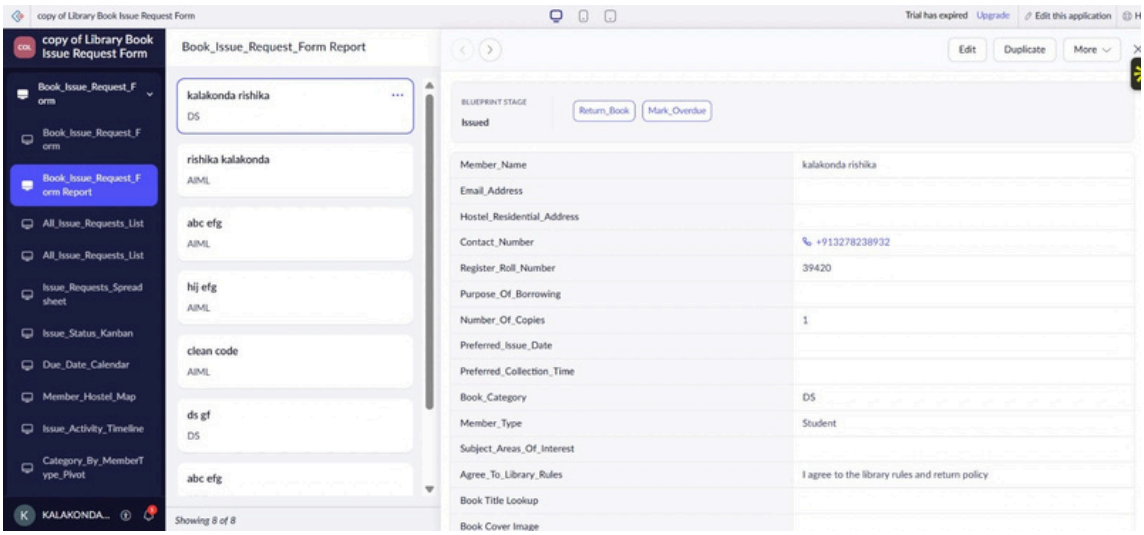
BLUEPRINT




copy of Library Book Issue Request Form
Book_Issue_Request_Form Report
kalakonda rishika
DS
rishika kalakonda
AIML
Edit
Duplicate
More
BLUEPRINT STAGE
Requested
Submit_Request
Member_Name
kalakonda rishika
Email_Address

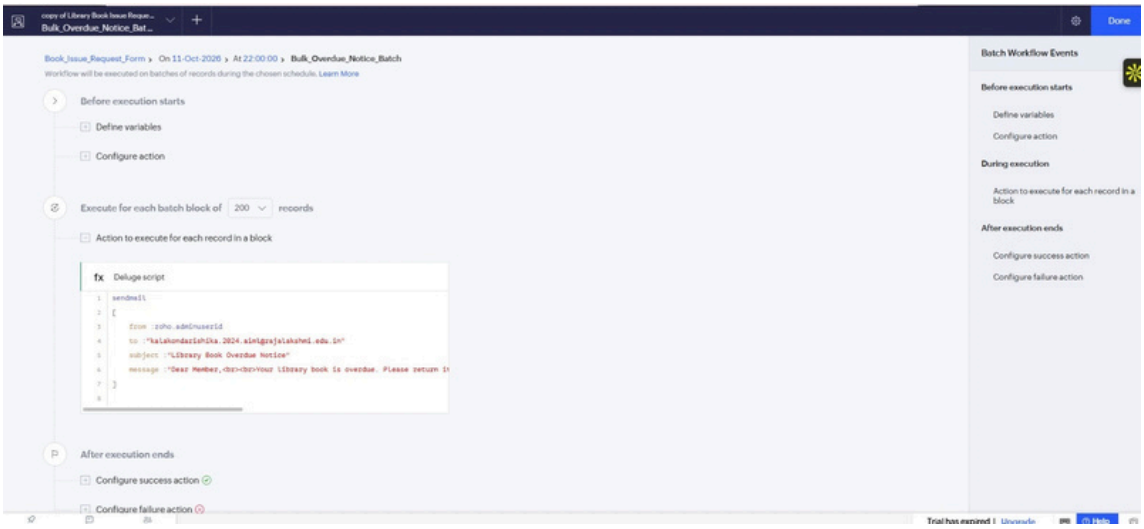

copy of Library Book Issue Request Form
Book_Issue_Request_Form Report
kalakonda rishika
DS
rishika kalakonda
Edit
Duplicate
More
BLUEPRINT STAGE
Approved
Issue_Book
Member_Name
kalakonda rishika

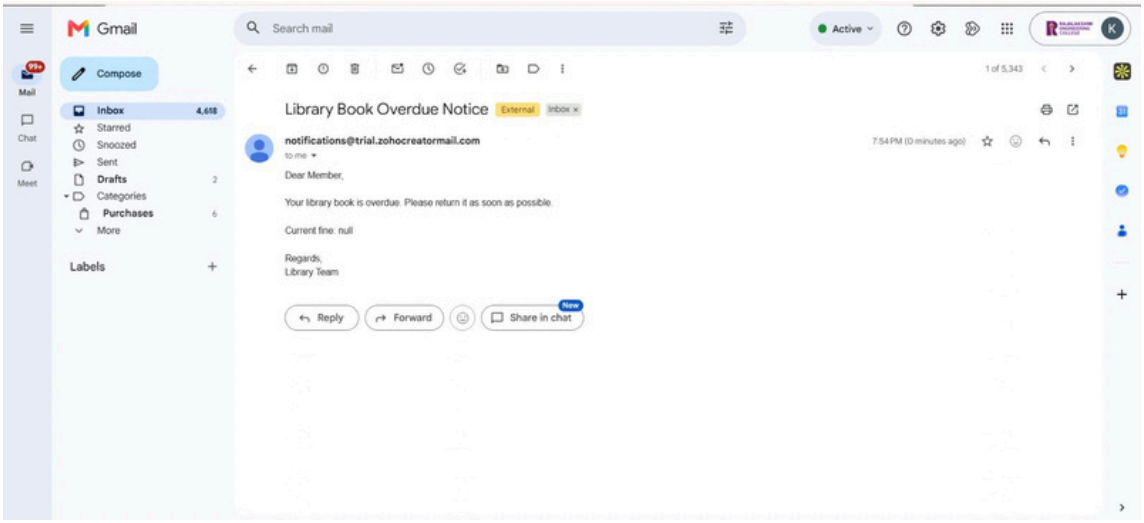

copy of Library Book Issue Request Form
Book_Issue_Request_Form Report
Are you sure you want to perform this transition?
Yes
No
kalakonda rishika
DS
rishika kalakonda
AIML
abc efg
AIML
hij efg
AIML
clean code
AIML
ds gf
DS
Contact_Number
+913278238932
Register_Roll_Number
39420
Number_Of_Copies
1
Book_Category
DS
Member_Type
Student


copy of Library Book Issue Request Form
Book_Issue_Request_Form Report
kalakonda rishika
DS
rishika kalakonda
AIML
abc efg
AIML
hij efg
AIML
clean code
AIML
ds gf
DS
abc efg
Showing 8 of 8
Edit
Duplicate
More
BLUEPRINT STAGE
Issued
Return_Book
Mark_Overdue
Member_Name
kalakonda rishika
Email_Address
Hostel_Residential_Address
Contact_Number
+913278238932
Register_Roll_Number
39420
Purpose_Of_Borrowing
Number_Of_Copies
1
Preferred_Issue_Date
Preferred_Collection_Time
Book_Category
DS
Member_Type
Student
Subject_Areas_Of_Interest
Agree_To_Library_Rules
I agree to the library rules and return policy
Book Title Lookup
Book Cover Image

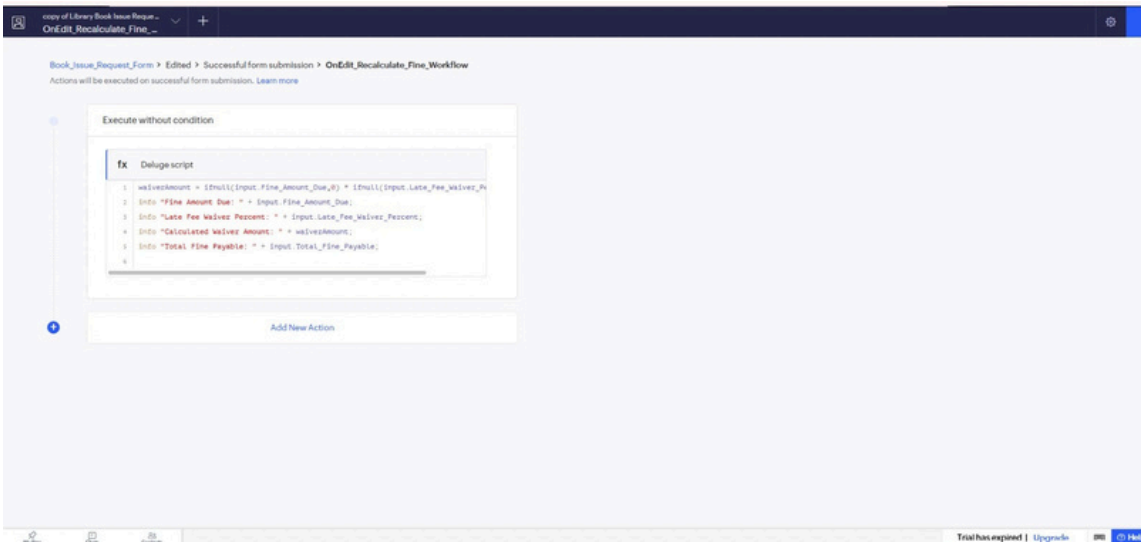

clean code
AIML
ds gf
Showing 9 of 9
Issue_status
Overdue
Due_date
12-Oct-2026
Book_Type
Regular

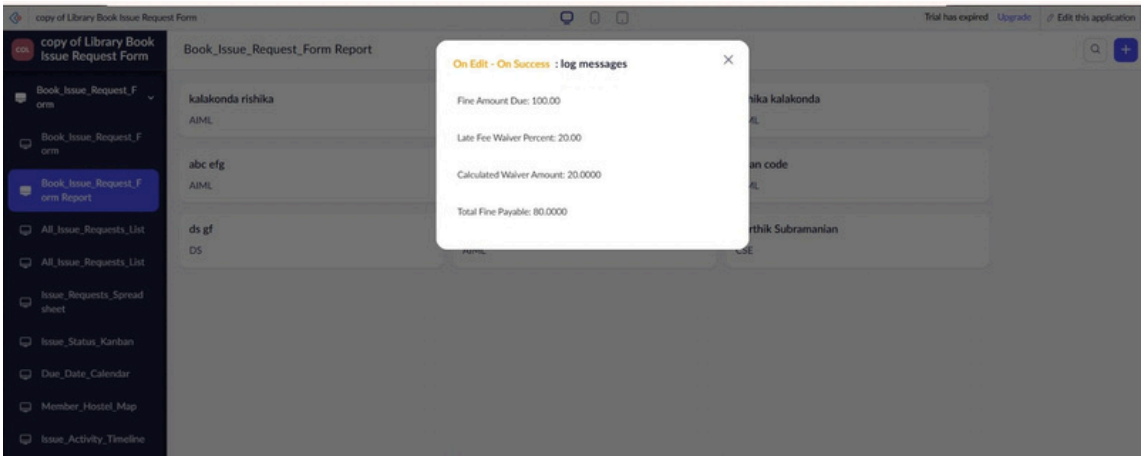
BATCH WORKFLOW



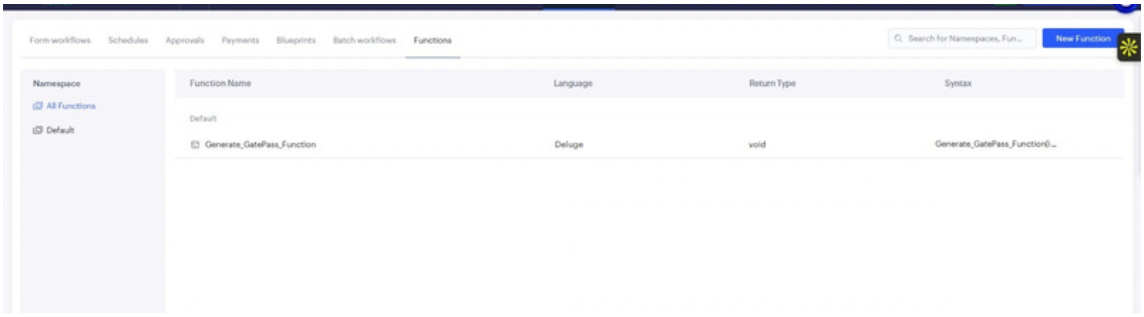


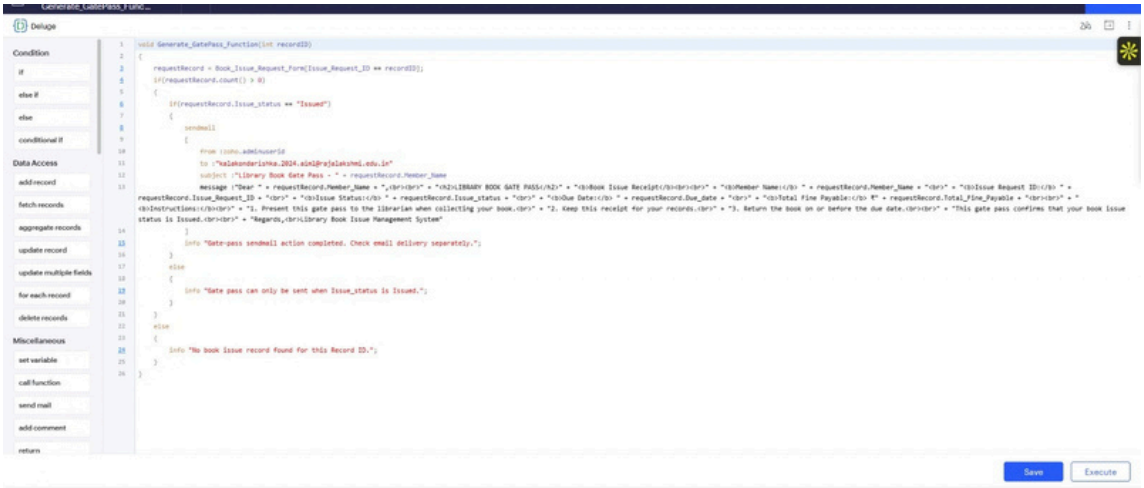
REPORT WORKFLOW

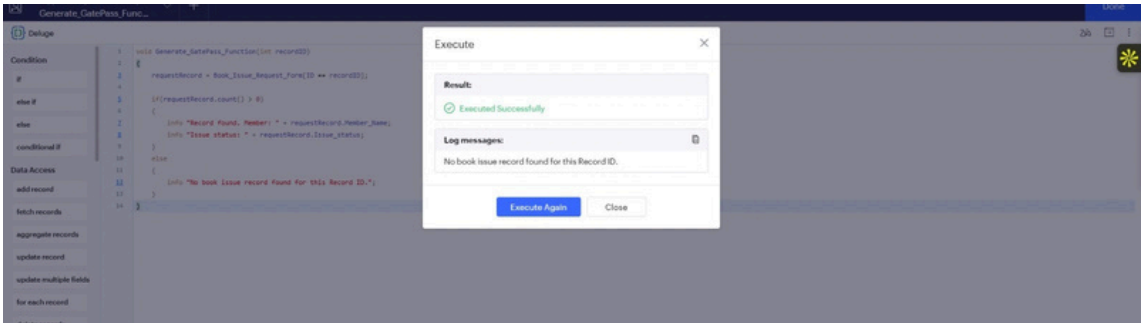


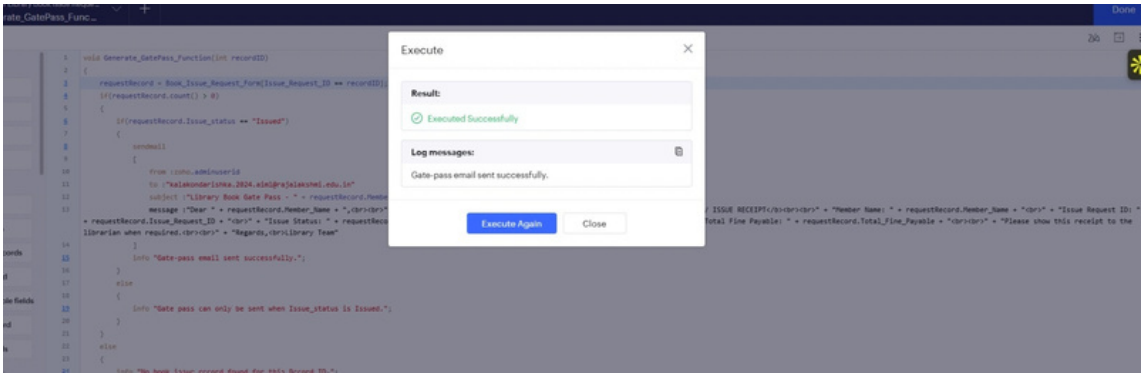

copy of Library Book Issue Request Form
Book_Issue_Request_Form Report
On Edit - On Success : log messages
Fine Amount Due: 100.00
Late Fee Waiver Percent: 20.00
Calculated Waiver Amount: 20.0000
Total Fine Payable: 80.0000
kalakonda rishika
AIML
abc efg
AIML
ds gf
DS

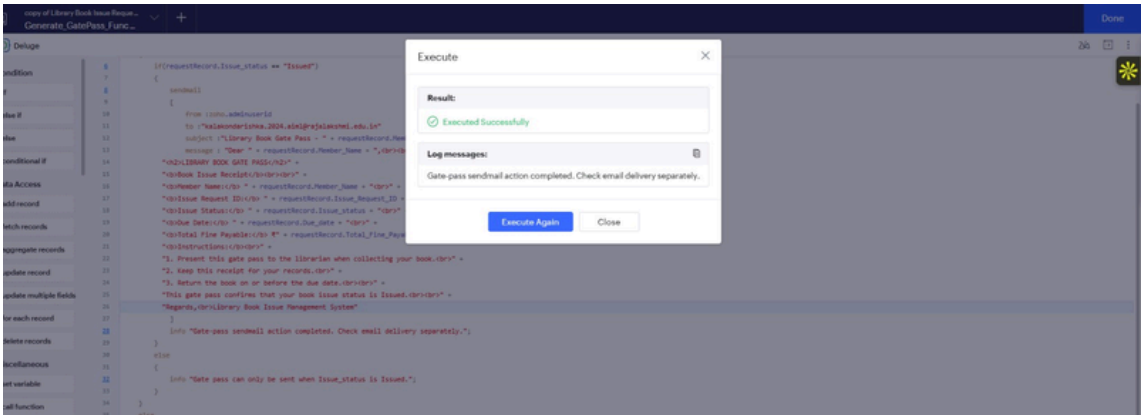
FUNCTIONS


Form workflows
Schedules
Approvals
Payments
Blueprints
Batch workflows
Functions
New Function
Namespace
All Functions
Default
Function Name
Language
Return Type
Syntax
Generate_GatePass_Function
Deluge
void


Deluge
Condition
Data Access
Miscellaneous
Save
Execute


Execute
Result:
Executed Successfully
Log messages:
No book issue record found for this Record ID.
Execute Again
Close





RESULT

The Library Book Issue Automation Suite was designed in Zoho Creator using Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows, and Deluge Functions. Fine calculation, Blueprint transitions, batch workflow configuration, and gate-pass function execution were demonstrated. These automations help streamline book requests, approvals, fine management, due-date reminders, overdue notifications, and gate-pass processing.